

Faults Along an Apollo Descent

The complete study — 180 injections, a relaxation ladder, and what separates a lost vehicle from a tight corridor

1 September 2026

Contents

1	Summary	1
2	The vehicle and the nominal	2
2.1	Where the initial state comes from	2
3	Study H — the injection campaign	2
4	The trajectories in three dimensions	5
5	Study H-R — is it the vehicle or the corridor?	5
5.1	The question	5
5.2	The ladder	6
5.3	What the ladder found	7
6	The last rung — everything dropped except $z > 0$	9
7	What the three campaigns mean together	9
8	Limitations	10
9	Reproducing	10

1 Summary

This document is three campaigns in one argument.

Study H flies one Apollo-anchored nominal descent and injects 12 engine plants at 15 points along it — 180 optimal-control solves — to ask *where on the descent is this fault survivable*. It lands 168 of them and finds 11 injections with no recovery trajectory at all.

Study H-R asks whether those failures are the vehicle or the problem statement, by re-solving each at four progressively weaker corridors. 3 of them recover — all at the first level, the glide cone alone.

The last rung, L4, drops every state constraint except $z > 0$: nothing is asked of the vehicle but to stay above the surface. 9 cases fail even there.

The three-line result:

Result	Count	Reading
Injections that land	168 of 180	Study H, inside the design corridor
Recoverable once the cone is relaxed	3	2 still land in the Apollo gate

Result	Count	Reading
Infeasible with only $z > 0$ enforced	9	resisted 22 seeds across four corridors

The headline correction is the last two lines together: Study H’s 11 “unrecoverable” injections are really **9** unrecoverable injections plus 3 that were fighting the glide cone.

2 The vehicle and the nominal

2.1 Where the initial state comes from

The anchor is the published Apollo approach-phase (P64) geometry rather than an invented state. Two points of that phase are well documented — high gate at 7,500 ft / 4.5 nmi / 500 ft/s, low gate at 500 ft / 2,000 ft / 60 ft/s — and the approach between them is flown essentially straight at the landing point. Interpolating along it to the speed this vehicle model admits with margin gives:

Anchor state	SI	Imperial
Altitude	736 m	2416 ft
Range to pad	2724 m	8937 ft
Horizontal speed	55.0 m/s	180 ft/s
Descent rate	15.6 m/s	51 ft/s
Line-of-sight depression	15.13°	
Flight-path angle	15.87°	

The last two lines are the consistency check: a vehicle flying the approach points its velocity vector at the landing site, and these agree to 0.75°. **This is a reconstruction of the geometry, not telemetry.**

Two properties of the vehicle matter for everything below. The engines sit at $y = \pm 0.25$ m, just inside the roll-trim limit $y_{eng} \leq dz_{eng} \tan \delta_{max} = 0.263$ m, which is what makes engine-out a fault with an answer rather than an automatic loss. And the glide cone is 12°, lowered from the earlier studies’ 30° because Apollo’s real approach is a ~15° path that a 30° cone forbids outright.

The Apollo-anchored nominal descent

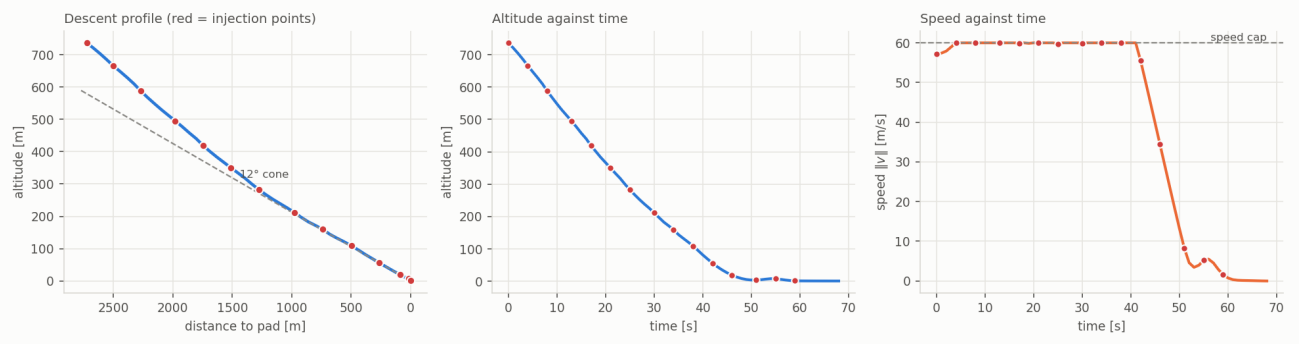


Figure 1: The Apollo-anchored nominal descent

The healthy vehicle reaches contact at $t = 65$ s with a gate margin of **0.60**.

3 Study H — the injection campaign

At each injection point the state is read straight off the nominal — all 22 of them — the plant becomes the damaged plant from that instant, and the vehicle re-plans over the time the nominal had left plus a reserve. The

healthy plant is run from every point as a control.

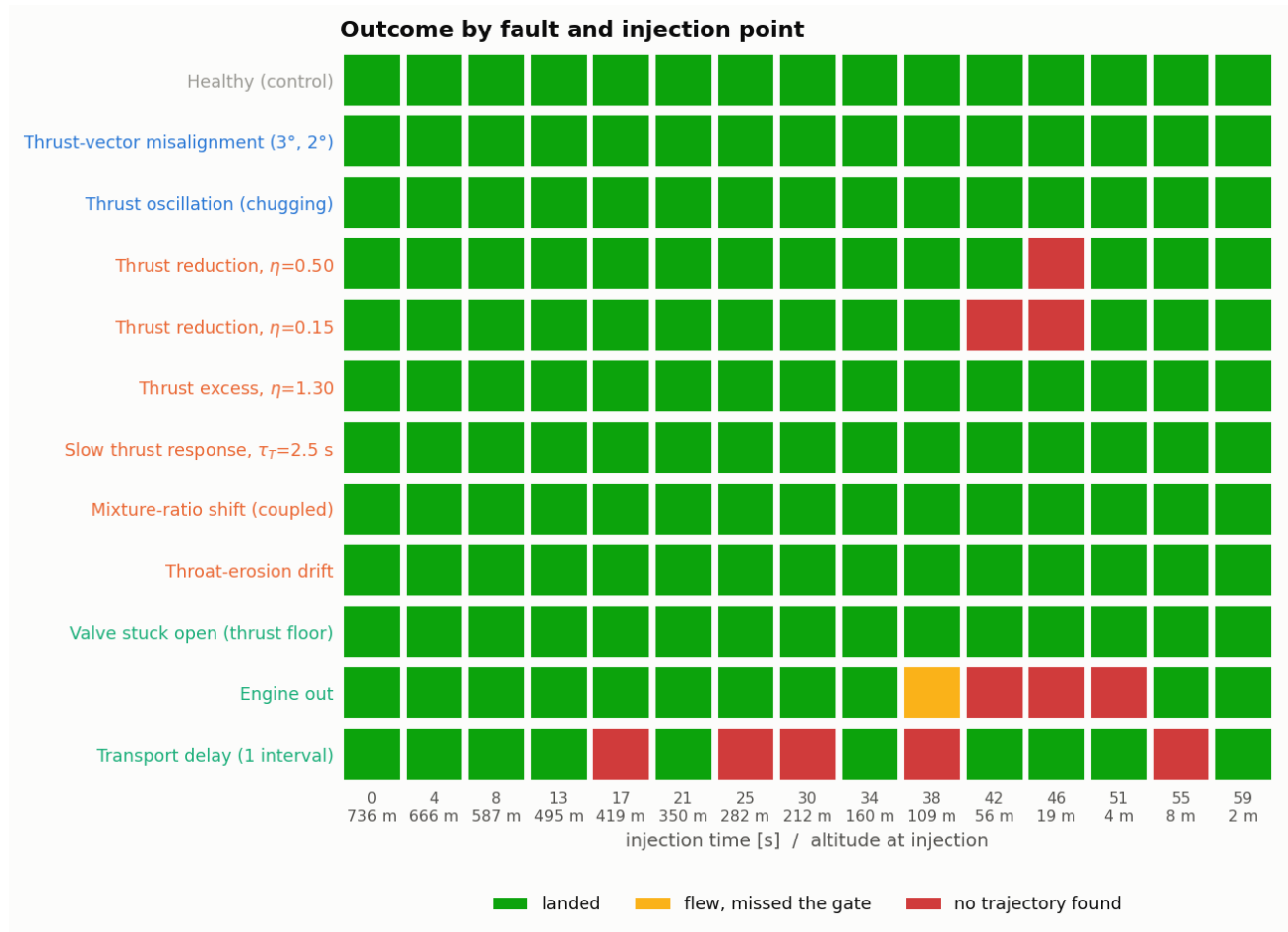


Figure 2: Outcome by fault and injection point

This is the study's overview figure, and the one worth reading first: one cell per solve, no averaging, time running left to right. Three groups fall out of it immediately.

Group	Faults	Which
Survive everywhere	7	TVC bias, chugging, $\eta=1.30$, $\tau_T=2.5$ s, mixture ratio, erosion drift, stuck open
Have a boundary	4	$\eta=0.50$, $\eta=0.15$, engine out, dead time
Never survive	0	—

The faults that fail do not fail randomly across the descent; they fail late. Every fault with a boundary has it in the braking phase, where the vehicle is low, still fast, and has committed its remaining altitude to the flare. A fault arriving during the cruise has hundreds of metres to trade for a re-plan; the same fault arriving during the flare has none. Altitude is the currency.

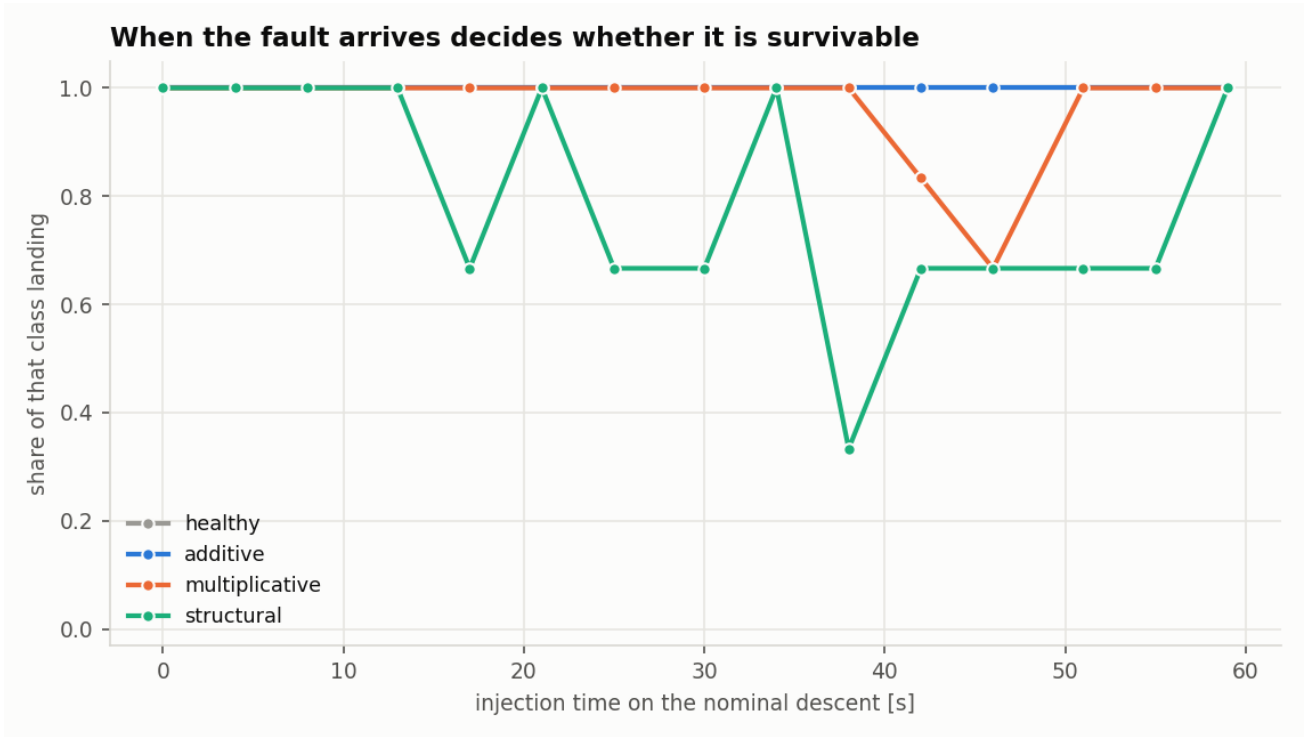


Figure 3: Landing share by fault class against injection time



Figure 4: Gate margin against injection time, per fault

4 The trajectories in three dimensions

Every figure so far collapses the path into the range-altitude plane — the plane the glide cone lives in, and therefore the plane the constraints are argued in. It is not the plane the vehicle flies in. A one-engine-out LM yaws under asymmetric thrust, the gimbal trims it, and what is left is a cross-range excursion that a 2-D plot draws as nothing at all.

Study H — every recovery, in three dimensions

180 injections, one panel per plant. Cross-range is invisible in the range-altitude plots; here it is not.

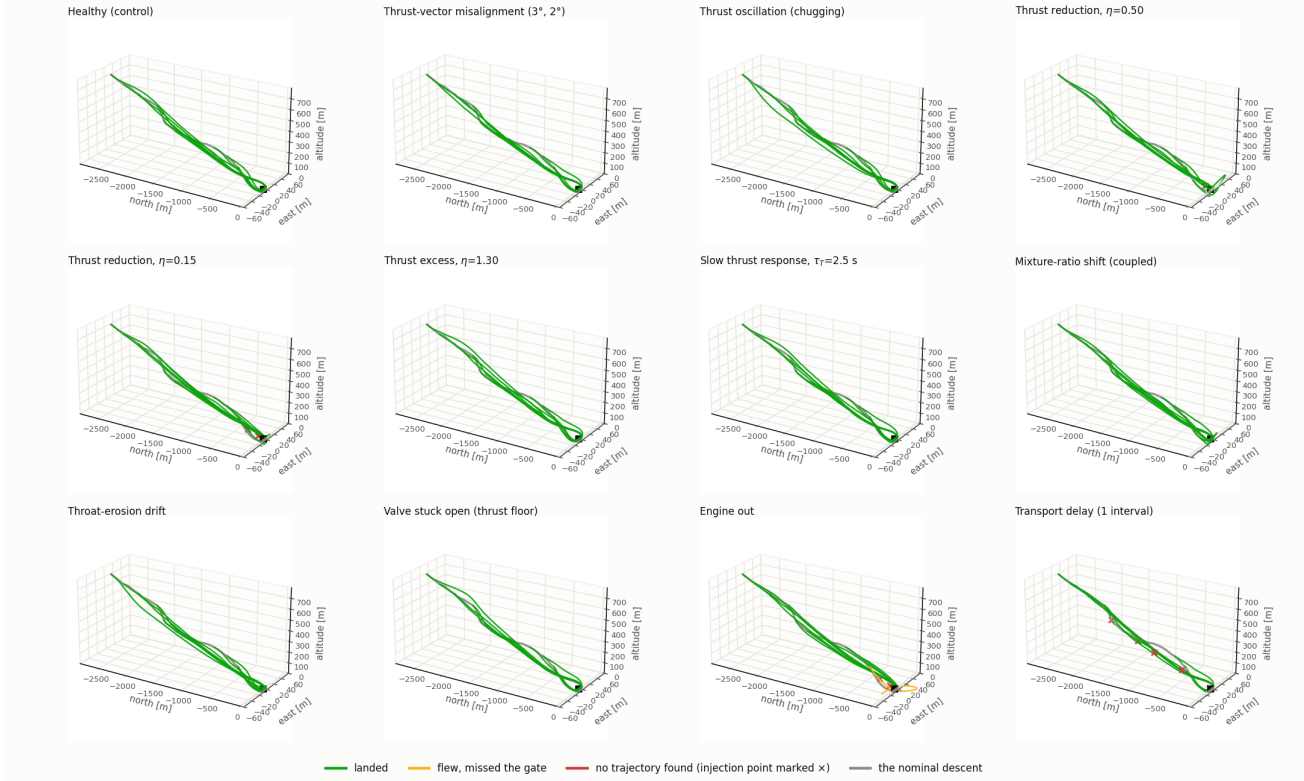


Figure 5: Study H — every recovery in 3-D, one panel per plant

The panels are worth comparing against each other rather than read one at a time. The thrust-magnitude faults stay in the vertical plane and differ from the nominal mostly in *when* they brake. The asymmetric ones — engine out above all — bulge sideways and come back, and the failures cluster at the end of the descent where there is no room left to do that.

5 Study H-R — is it the vehicle or the corridor?

5.1 The question

The problem those failed solves could not satisfy carries a corridor of path constraints, and not all of them are physics:

Constraint	Study H value	What it actually is
Glide cone	12° above horizontal	A scenario choice — it rules out the dive-and-crawl-back trajectory family, not something the vehicle cannot fly.
Attitude	45° per axis	Planner comfort. <code>hard_loss()</code> puts real loss of control at 90°.
Body rate	10°/s	Comfort again; <code>hard_loss()</code> uses 120°/s.

Constraint	Study H value	What it actually is
Speed	60 m/s per axis and as a norm	The model has no aerodynamic or structural speed limit at all.

So “no trajectory exists” may mean “no trajectory exists **that stays inside the corridor**” — a very different engineering statement.

5.2 The ladder

Level	Corridor	Constraints removed outright
L1	glide cone 12 to 6 deg	none dropped
L2	+ attitude 45 to 60 deg, rate 10 to 20 deg/s	none dropped
L3	+ speed 60 to 90 m/s (axis and norm)	none dropped
L4	all state constraints dropped except $z > 0$	cone, vel, att, rate

Thrust and gimbal bounds stand at every level — they are hardware. The Apollo gate stands at every level: a relaxed solve still has to touch down inside the same gate to count as a landing. And the altitude floor stands at every level, so $z > 0$ holds throughout — L4 leaves it as the *only* constraint on the state.

Effort is matched deliberately. Study H spent 7 seeds at up to 1,200 iterations on each of these cases; each rung adds 3 horizon seeds at 1,200, and the terminal rung L4 — where a failure is the study’s strongest claim — adds 6. A case still infeasible there has resisted **22 seeds**.

5.3 What the ladder found

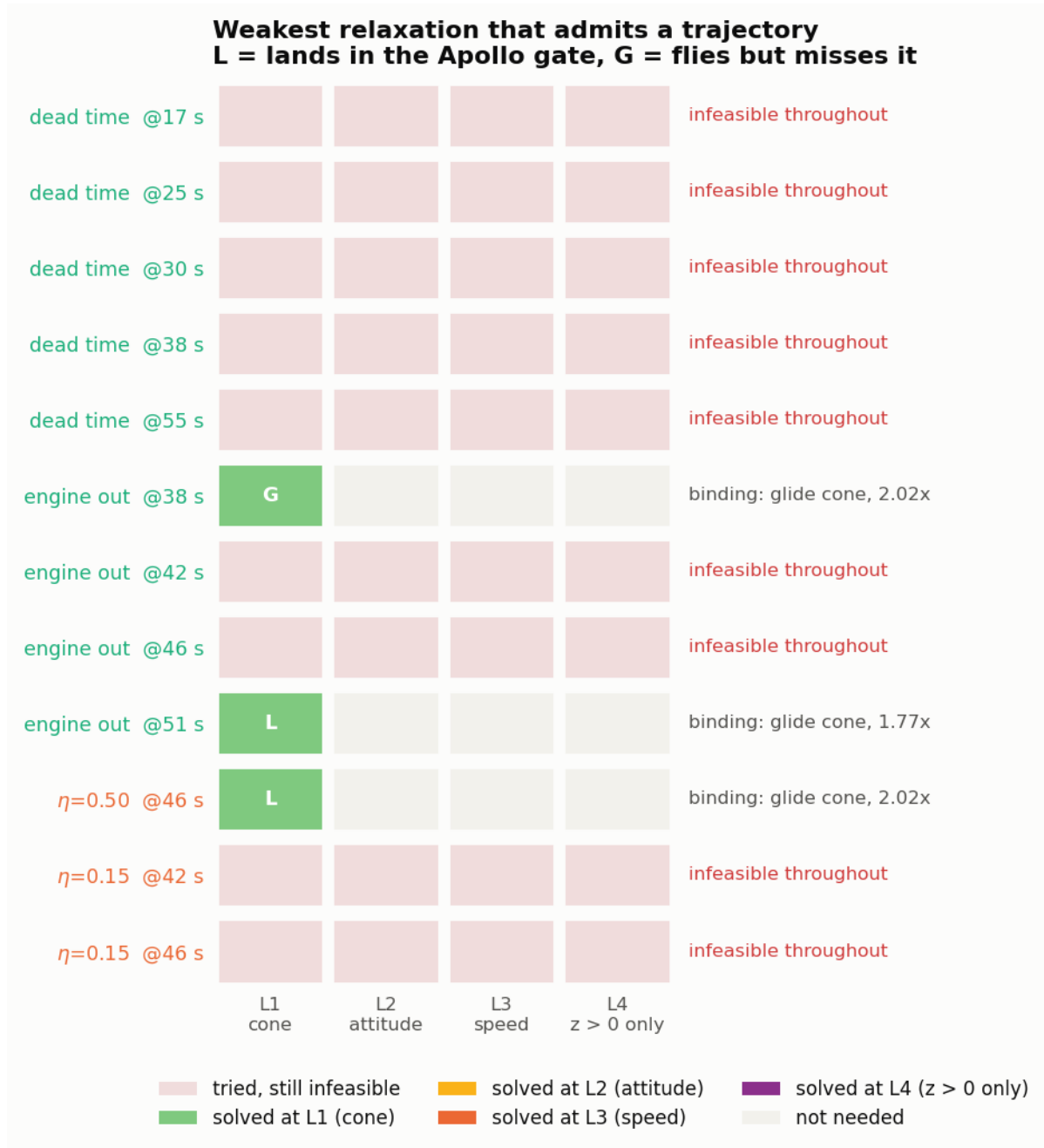


Figure 6: The weakest relaxation that admits a trajectory

Fault	Injected	Altitude	Study H	Solved at	Outcome	Gate margin	Binding baseline limit
dead time	17 s	419 m	no recovery	—	no recovery	—	n/a
dead time	25 s	282 m	no recovery	—	no recovery	—	n/a
dead time	30 s	212 m	no recovery	—	no recovery	—	n/a
dead time	38 s	109 m	no recovery	—	no recovery	—	n/a
dead time	55 s	8 m	no recovery	—	no recovery	—	n/a
engine out	38 s	109 m	gate miss	L1	gate miss	6.88	glide cone

Fault	Injected	Altitude	Study H	Solved at	Outcome	Gate margin	Binding baseline limit
engine out	42 s	56 m	no recovery	—	no recovery	—	n/a
engine out	46 s	19 m	no recovery	—	no recovery	—	n/a
engine out	51 s	4 m	no recovery	L1	land	0.60	glide cone
$\eta=0.50$	46 s	19 m	no recovery	L1	land	0.60	glide cone
$\eta=0.15$	42 s	56 m	no recovery	—	no recovery	—	n/a
$\eta=0.15$	46 s	19 m	no recovery	—	no recovery	—	n/a

Everything that recovered, recovered at L1. The attitude, rate and speed relaxations unlocked nothing at all: no case anywhere in this study was limited by how hard it was allowed to pitch, spin or fly. The single binding constraint in the whole campaign is the glide cone, and the recoveries that need it fly a 6.0–6.8° path against a 12° floor. They are *shallower* approaches, not more violent ones.

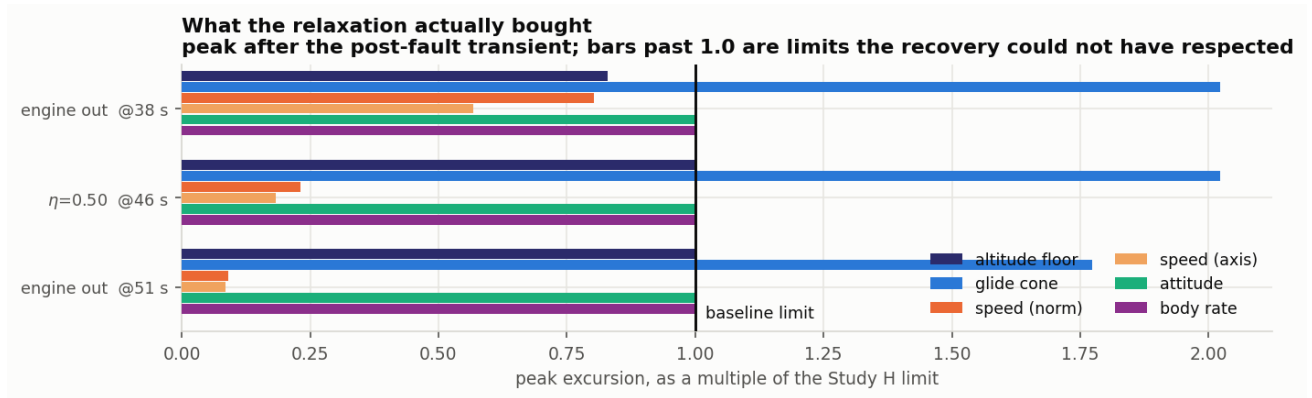


Figure 7: Peak excursion beyond each baseline limit

These excursions are measured **after the post-fault transient**, from node 6 onward. `solve_ocp` opens a corridor over the first 6 nodes — up to $3\times$ the rate limit — that shrinks to the nominal envelope, and the baseline problem opens it too. An excursion inside that window is something Study H already permitted, so counting it would credit the recovery to the wrong constraint: measured over the whole trajectory two of these recoveries look bought by the body-rate limit at $\sim 2\times$; measured after the ramp their rate and attitude peaks are exactly $1.00\times$ and only the cone is broken.

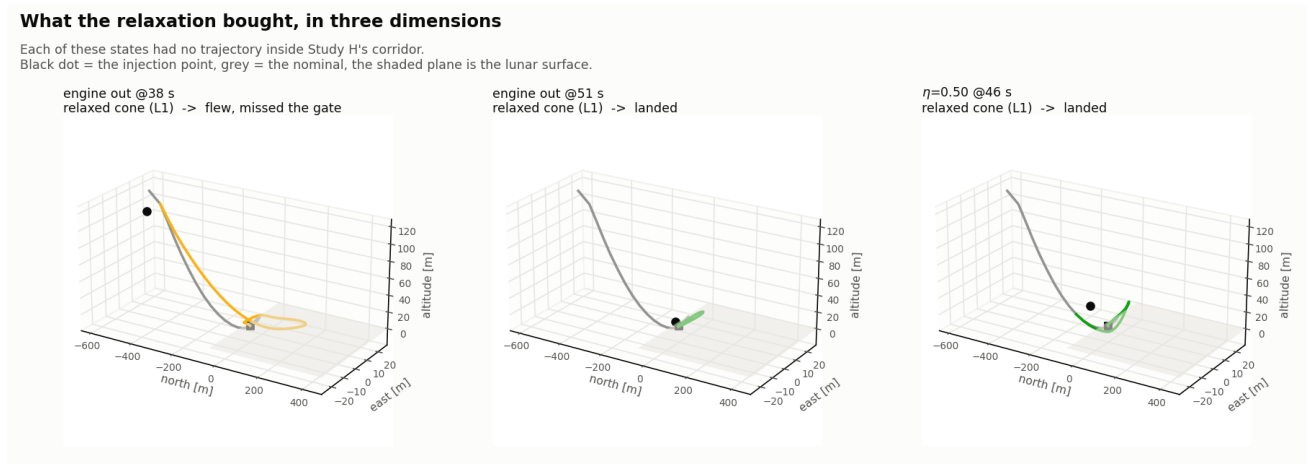


Figure 8: What the relaxation bought, in 3-D

Where the losses sit on the descent

9 injection points still have no trajectory with every state constraint dropped except $z > 0$ (red x); 3 recovered once the glide cone was opened (green ▲). Green dots are the injection points that landed inside Study H's own corridor. The losses are not spread along the descent — they are stacked at the end of it.

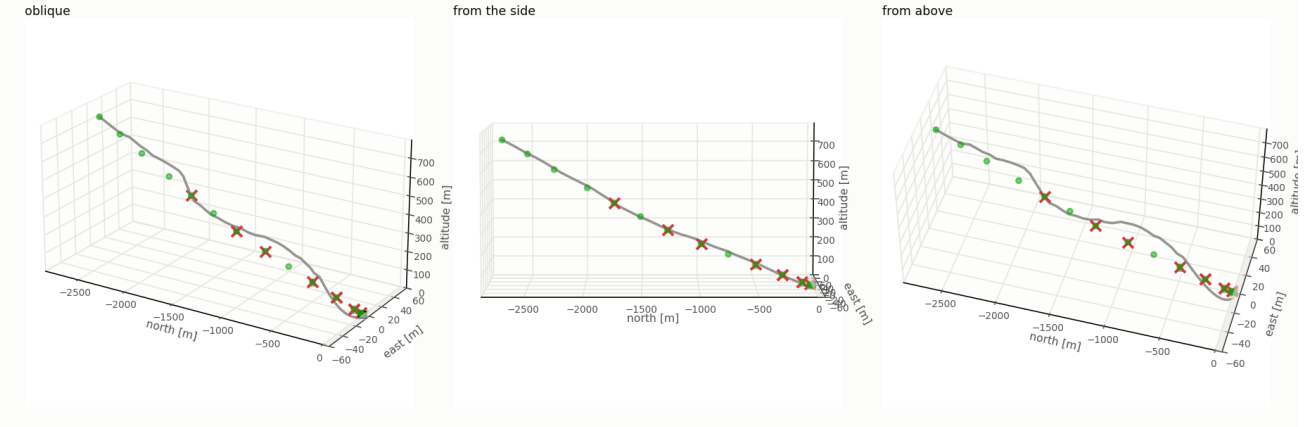


Figure 9: Where the surviving losses sit on the descent

6 The last rung — everything dropped except $z > 0$

L4 is the weakest problem in this study. The glide cone, the speed box, the attitude limits and the rate limits are all gone; the only thing still asked of the state is that the vehicle stays above the lunar surface.

The altitude floor is never dropped, at any level. It is worth saying why, because dropping it looks superficially like the natural end of a relaxation ladder. It is not — the landing gate inspects only the *touchdown state*, so a path routed through the ground can return to the pad with a clean gate margin and be scored as a landing. That is an arithmetic result rather than a flight one, and admitting it would corrupt the very count this study exists to produce. $z > 0$ is the boundary between a relaxed problem and a meaningless one.

9 of the 11 cases fail even here, with nothing in their way but the ground. They are short of control authority outright: no guidance law, however permissive about attitude, rate, speed or approach angle, recovers them. These are the study's real losses.

7 What the three campaigns mean together

A solver's "infeasible" is a property of the constraint set, and must be reported as one. 11 failures became 9 once the same states and the same plants were given a wider corridor. Any study that reports fault survivability from a constrained planner is reporting its constraints as much as its vehicle, and the only way to know the split is to vary them deliberately.

Severity should be quoted with the corridor attached. "Unrecoverable after $t = X$ " is incomplete. The defensible form is "unrecoverable after $t = X$ inside a 12° cone, and after $t = Y$ with the cone opened to 6° " — and the gap between the two numbers is exactly the value of an envelope-expanding guidance mode.

The cone is doing more work than any other constraint. It was chosen for a near-vertical scenario, inherited into a shallow Apollo approach, and it is the single limit that decides 3 of these cases. Nothing else in the corridor mattered anywhere in the campaign. If one number in this problem statement deserves re-derivation from vehicle and mission requirements rather than convenience, it is that one.

Altitude is the currency, and the boundary is a time. Every fault with a boundary has it late, in the braking phase. The single number a fault-tolerance budget should carry is the injection time past which recovery stops existing — and after this study, two of them: one for the design corridor and one for the open one.

9 genuine losses, not 11. After four corridors and 22 seeds, the failures that remain are $\eta=0.15$, dead time, engine out — and all five dead time injections are among them. That is a smaller and much better-defended

claim than the campaign originally supported.

8 Limitations

- **Local infeasibility is not global.** 22 seeds is strong evidence, not proof.
- **Instantaneous, perfect detection.** No isolation, identification or reaction delay; every number here is an upper bound. Studies D and E measure the gap.
- **Open loop.** Each result is one solve, not a closed loop re-solving as it flies.
- **One nominal.** All 180 solves depart from a single reference trajectory; a steeper or faster approach would move the boundaries.
- **The relaxed limits are round numbers,** chosen to bracket the baseline rather than derived from vehicle structure. The ladder localises which constraint binds, not the exact threshold.
- **Constant mass,** inherited from the base model, understating the propellant cost of a partial fault.
- **15 injection points** resolve a boundary to about 4 s.

9 Reproducing

```
python studies/fault_injection/apollo_nominal.py           # the reference descent
python studies/fault_injection/run_injection_study.py      # Study H campaign
python studies/fault_injection/harden.py                  # H hardening pass
python studies/fault_injection/analyse_injection.py        # H figures + JSON
python studies/fault_injection/run_relaxed_study.py        # H-R ladder, L1-L4
python studies/fault_injection/analyse_relaxed.py          # H-R figures + JSON
python studies/fault_injection/plot_trajectories_3d.py     # the 3-D figures
python studies/fault_injection/build_combined_report.py    # this document
```